

Evolution of Deep Learning Models for Plant Disease Detection

Team Members:

Member 1 -Rupan Bittu Roll No:142301019

Member 2 -Chittiboina arun Shyam Roll No:142301007

Member 3 -Dadi lakshmi Harsha Roll No:142301008

Member 4 -Durga Sri R Roll No:142202001

April 2026

1 Introduction

This project focuses on automated plant disease detection across 38 classes using the PlantVillage dataset. The approach evolves from simple convolutional models to advanced architectures such as EfficientNet, ResNet with Grad-CAM, and Vision Transformers.

2 Phase 1: Baseline Simple CNN

The initial phase involved building a custom **SimpleCNN** model.

- **Architecture:** Convolutional layers with ReLU activation and Max-Pooling, followed by fully connected layers.
 - **Batch Normalization:** Used to stabilize training by normalizing activations across mini-batches.
 - **Advantages:** Fast training and low computational cost.
 - **Limitation:** Limited ability to capture fine-grained disease patterns.
-

3 Phase 2: EfficientNet Model

To improve efficiency and performance, EfficientNet architecture was introduced.

- **Architecture:** EfficientNet uses compound scaling to uniformly scale depth, width, and resolution of the network.
- **Key Idea:** Instead of arbitrarily increasing layers, EfficientNet balances:
 - Network depth (number of layers)
 - Network width (number of channels)
 - Image resolution
- **Transfer Learning:** Pretrained EfficientNet weights were used to leverage learned visual features.
- **Advantages:**
 - Better accuracy with fewer parameters

- Faster training compared to deeper CNNs like ResNet
 - Efficient memory usage
 - **Observation:** EfficientNet improved performance over the baseline CNN while maintaining computational efficiency.
-

4 Phase 3: ResNet + Grad-CAM Model

To further enhance performance and interpretability, a ResNet-based model with Grad-CAM was implemented.

- **Transfer Learning:** Pretrained ResNet50 model used for feature extraction.
 - **Grad-CAM:** Provides visual explanations by highlighting important regions in the image.
 - **Gradient Clipping:** Prevents exploding gradients by limiting gradient magnitude during training.
 - **Learning Rate Scheduler:** Dynamically adjusts learning rate to improve convergence.
 - **Performance:**
 - Validation Accuracy: Up to 98.98%
 - Test Accuracy: Up to 99.16%
 - **Significance:** Combines high accuracy with interpretability, making predictions explainable.
-

5 Phase 4: Vision Transformer (ViT)

The final phase explored transformer-based architectures.

- **Architecture:** Images are divided into patches and processed using self-attention.
 - **Self-Attention:** Captures global relationships across the entire image.
 - **Hyperparameter Tuning:** Included weight decay and gradient clipping.
 - **Observation:** Slightly lower accuracy but better generalization capability.
-

6 Training Techniques Used

- **Batch Normalization:** Normalizes activations to improve training stability and convergence speed.
 - **Gradient Clipping:** Limits gradient values to prevent instability during backpropagation.
 - **Learning Rate Scheduler:** Adjusts learning rate dynamically to improve optimization and convergence.
 - **Dropout:** Reduces overfitting by randomly disabling neurons during training.
-

Table 1: Comparative Model Metrics

Model Type	Best LR	Weight Decay	Val Acc	Test Acc
EfficientNet	0.001	0.0005	High	High
Grad-CAM ResNet	0.0005	0.001	98.93%	99.16%
Vision Transformer	0.0001	0.100	95.44%	95.54%

7 Model Comparison

—

8 Conclusion

The project demonstrates a progressive improvement in model performance from simple CNNs to advanced architectures.

EfficientNet provided a balance between performance and efficiency, while the ResNet + Grad-CAM model achieved the highest accuracy with interpretability. Vision Transformers contributed strong generalization capabilities.

Overall, the study highlights the importance of architectural design, optimization strategies, and explainability in deep learning systems.